# Experiment 2

Suppose $\lim_{K \to \infty} x[K] = C_A 1$ → Protocol A convergence coordinate

↳ The robots eventually converge to one common x-coordinate denoted $C_A$

Then,

$$V^T x[0] = V^T (C_A 1)$$

This is because the weighted sum is conserved, staying constant regardless of position, as shown in part c of the checkpoint.

$$V^T x[0] = C_A V^T 1$$

Since $C_A$ is a scalar. Thus,

$$C_{Ax} = \frac{V^T x[0]}{V^T 1} \quad \text{and similarly for } y \quad C_{Ay} = \frac{V^T y[0]}{V^T 1}$$

Protocol A converges to the degree-weighted average of the initial positions

Suppose $\lim_{k\to\infty} x[k] = C_B 1$ → Protocol B's convergence coordinate

Then,

$$1^T x[0] = 1^T (C_B 1)$$

This is because the ordinary sum of robot coordinates is preserved for protocol B.

$$1^T x[0] = C_B 1^T 1 = C_B \cdot n$$

where $n$ is the number of robots

$$C_{Bx} = \frac{1^T x[0]}{n}, \quad C_{By} = \frac{1^T y[0]}{n}$$

Protocol B converges to the ordinary average of the initial positions

For testing the formulas, $g_A = \frac{V^T P \theta}{V^T 1}$ (↓ Predicted A), $g_B = \frac{1^T P \theta}{n}$ (↓ Predicted B)